

National Cybersecurity Authority (NCA) Essential Cybersecurity Controls (ECC)

NCS ECC framework summary

Saudi Arabia's Vision for 2030 aims for comprehensive national improvement, focusing on security, economy, and citizens' well-being. Emphasizing digital transformation, the Vision seeks to enhance digital infrastructure to align with global advancements in IT/OT systems, data management, and artificial intelligence. The National Cybersecurity Authority (NCA) was established to safeguard vital interests, national security, and critical infrastructures.

NCA develops Essential Cybersecurity Controls (ECC) to set minimum requirements for organizations, ensuring compliance and continuous monitoring. All organizations must adhere to ECC mandates outlined in Royal Decrees to enhance cybersecurity resilience.

The controls aim to establish minimum cybersecurity standards for organizations' IT assets, drawing from industry best practices to mitigate internal and external threats. The essential cybersecurity controls consist of 5 main domains, 29 subdomains, and a 114 cybersecurity controls.

Darktrace and ECC: Framework Applicability

Darktrace can aid organizations in aligning with Saudi Arabia's Essential Cybersecurity Controls (ECC) as mandated by the National Cybersecurity Authority (NCA).

By leveraging a combination of AI methods to learn from your unique business data, Darktrace continuously monitors, autonomously responds to, and provides risk management for both internal and external threats in real-time, continuously monitoring without any additional risk induced by external connectivity.

Darktrace is the only cybersecurity solution that natively covers IT and OT providing visibility of OT, IoT, and IT assets in unison encompassing network and cloud-connected IT systems to specialized OT assets, achieving greater visibility of OT and IT devices across all levels of the Purdue Model.

Darktrace aligns with the emphasis on protecting critical infrastructures through proactive threat detection and adaptive defense mechanisms.

ECC domains and structure



Domain No.	Domain	Subdomain No.	Subdomain	Control No.	Control	Darktrace Coverage
1	Cybersecurity Governance	1-1	Cybersecurity Strategy	1-1-2	A roadmap must be executed to implement the cybersecurity strategy.	Darktrace provides visibility into existing risks which can be used when prioritizing projects in a roadmap. Additionally, Darktrace can help organizations understand how prepared they are for responding to an incident and whether response capabilities should be prioritized in a roadmap.
1	Cybersecurity Governance	1-3	Cybersecurity Policies and Procedures	1-3-2	The cybersecurity function must ensure that the cybersecurity policies and procedures are implemented.	Darktrace can identify and notify security teams of policy violations and also interrupt the offending activity thus enforcing policy compliance.
1	Cybersecurity Governance	1-5	Cybersecurity Risk Management	1-5-2	The cybersecurity risk management methodology and procedures must be implemented by the cybersecurity function.	Darktrace can be configured to support an organization's governance and risk management processes with varying levels of automation and/or alerting.
1	Cybersecurity Governance	1-6	Cybersecurity in Information and Technology Project Management	1-6-2	The cybersecurity requirements in project and assets (information/technology) change management must include at least the following: 1-6-2-1 Vulnerability assessment and remediation. 1-6-2-2 Conducting a configurations' review, secure configuration and hardening and patching before changes or going live for technology projects.	Darktrace can highlight misconfigurations across application architectures and also provide references for best practices to assist in remediation.
1	Cybersecurity Governance	1-6	Cybersecurity in Information and Technology Project Management	1-6-3	The cybersecurity requirements related to software and application development projects must include at least the following: 1-6-3-1 Using secure coding standards. 1-6-3-2 Using trusted and licensed sources for software development tools and libraries. 1-6-3-3 Conducting compliance test for software against the defined organizational cybersecurity requirements. 1-6-3-4 Secure integration between software components. 1-6-3-5 Conducting a configurations' review, secure configuration and hardening and patching before going live for software products.	Darktrace can highlight misconfigurations across application architectures and also provide references for best practices to assist in remediation.
1	Cybersecurity Governance	1-7	Compliance with Cybersecurity Standards, Laws and Regulations	1-7-1	The organization must comply with related national cybersecurity laws and regulations.	Darktrace helps organizations manage legal and regulatory requirements through both its default functionality and by providing configurability for organizations to specify their own controls.
1	Cybersecurity Governance	1-7	Compliance with Cybersecurity Standards, Laws and Regulations	1-7-2	The organization must comply with any nationally-approved international agreements and commitments related to cybersecurity.	Darktrace helps organizations manage legal and regulatory requirements through both its default functionality and by providing configurability for organizations to specify their own controls.
1	Cybersecurity Governance	1-8	Periodical Cybersecurity Review and Audit	1-8-1	Cybersecurity reviews must be conducted periodically by the cybersecurity function in the organization to assess the compliance with the cybersecurity controls in the organization.	Darktrace assists in cybersecurity reviews by providing security teams with visibility into actual and potential activity within the organization. Darktrace can also help enforce compliance through realtime alerting and disruption of non-compliant activities.

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1	Cybersecurity Governance	1-9	Cybersecurity in Human Resources	1-9-5	Personnel access to information and technology assets must be reviewed and removed immediately upon termination/separation.	Darktrace can alert on and disrupt activities related to personnel whose access should have been removed. Additionally, Darktrace can be configured to increase sensitivity on assets related to personnel with nearing termination dates so that anomalous actions are alerted on and disrupted more aggressively.
2	Cybersecurity Defense	2-1	Asset Management	2-1-2	The cybersecurity requirements for managing information and technology assets must be implemented.	Darktrace provides discovery of information assets, real-time visibility of activity, and the capability to alert on and disrupt activities based on asset classification.
2	Cybersecurity Defense	2-1	Asset Management	2-1-4	Acceptable use policy of information and technology assets must be implemented.	Darktrace continuously monitors the digital activity within an organization and can be configured to interrupt actions that violate acceptable use policies.
2	Cybersecurity Defense	2-1	Asset Management	2-1-5	Information and technology assets must be classified, labeled and handled as per related law and regulatory requirements.	Darktrace inventories assets seen communicating on the network and can also discover and inventory externally facing assets.
2	Cybersecurity Defense	2-2	Identity and Access Management	2-2-2	The cybersecurity requirements for identity and access management must be implemented.	Darktrace provides visibility into the actions of identities and credentials within an organization and can take action when anomalous behavior is detected.
2	Cybersecurity Defense	2-2	Identity and Access Management	2-2-3	The cybersecurity requirements for identity and access management must include at least the following 2-2-3-1 User authentication based on username and password. 2-2-3-2 Multi-factor authentication for remote access. 2-2-3-3 User authorization based on identity and access control principles: Need-to-Know and Need-to-Use, Least Privilege and Segregation of Duties. 2-2-3-4 Privileged access management. 2-2-3-5 Periodic review of users' identities and access rights.	Darktrace provides visibility into the actions of identities and credentials within an organization and can take action when anomalous behavior is detected.
2	Cybersecurity Defense	2-3	Information System and Information Processing Facilities Protection	2-3-2	The cybersecurity requirements for protecting information systems and information processing facilities must be implemented.	Darktrace continuously monitors information technology systems, and can identify potential vulnerabilities, non-compliant actions, and anomalous activity, and protect systems in real-time with respond capabilities.
2	Cybersecurity Defense	2-3	Information System and Information Processing Facilities Protection	2-3-3	The cybersecurity requirements for protecting information systems and information processing facilities must include at least the following: 2-3-3-1 Advanced, up-to-date and secure management of malware and virus protection on servers and workstations. 2-3-3-2 Restricted use and secure handling of external storage media. 2-3-3-3 Patch management for information systems, software and devices. 2-3-3-4 Centralized clock synchronization with an accurate and trusted source (e.g., Saudi Standards, Metrology and Quality Organization (SASO)).	Darktrace continuously monitors information technology systems, and can identify potential vulnerabilities, non-compliant actions, and anomalous activity, and protect systems in real-time with respond capabilities.

Domain No.	Domain	Subdomain No.	Subdomain	Control No.	Control	Darktrace Coverage
2	Cybersecurity Defense	2-4	Email Protection	2-4-2	The cybersecurity requirements for email service must be implemented.	Darktrace provides advanced email protection including phishing, business email compromise, malicious content, misdirected recipients, and graymail.
2	Cybersecurity Defense	2-4	Email Protection	2-4-3	The cybersecurity requirements for protecting the email service must include at the least the following: 2-4-3-1 Analyzing and filtering email messages (specifically phishing emails and spam) using advanced and up-to-date email protection techniques. 2-4-3-2 Multi-factor authentication for remote and webmail access to email service. 2-4-3-3 Email archiving and backup. 2-4-3-4 Secure management and protection against Advanced Persistent Threats (APT), which normally utilize zero-day viruses and malware. 2-4-3-5 Validation of the organization's email service domains (e.g., using Sender Policy Framework (SPF)).	Darktrace provides advanced email protection including phishing, business email compromise, malicious content, misdirected recipients, and graymail.
2	Cybersecurity Defense	2-5	Networks Security Management	2-5-2	The cybersecurity requirements for network security management must be implemented.	Darktrace continuously monitors network activity and identifies and disrupts anomalous and non-compliant events. Additionally, Darktrace continuously analyzes potential traffic and can alert on increased risks and potential non-compliant configurations.
2	Cybersecurity Defense	2-5	Networks Security Management	2-5-3	The cybersecurity requirements for network security management must include at least the following: 2-5-3-1 Logical or physical segregation and segmentation of network segments using firewalls and defense-in-depth principles. 2-5-3-2 Network segregation between production, test and development environments. 2-5-3-3 Secure browsing and Internet connectivity including restrictions on the use of file storage/sharing and remote access websites, and protection against suspicious websites. 2-5-3-4 Wireless network protection using strong authentication and encryption techniques. A comprehensive risk assessment and management exercise must be conducted to assess and manage the cyber risks prior to connecting any wireless networks to the organization's internal network. 2-5-3-5 Management and restrictions on network services, protocols and ports. 2-5-3-6 Intrusion Prevention Systems (IPS). 2-5-3-7 Security of Domain Name Service (DNS). 2-5-3-8 Secure management and protection of Internet browsing channel against Advanced Persistent Threats (APT), which normally utilize zero-day viruses and malware.	Darktrace continuously monitors network activity and identifies and disrupts anomalous and non-compliant events. Additionally, Darktrace continuously analyzes potential traffic and can alert on increased risks and potential non-compliant configurations.
2	Cybersecurity Defense	2-7	Data and Information Protection	2-7-2	The cybersecurity requirements for protecting and handling data and information must be implemented.	Darktrace can be configured to provide more sensitive protection to assets containing sensitive data in order to more aggressively identify and interrupt anomalous activity.

Domain No.	Domain	Subdomain No.	Subdomain	Control No.	Control	Darktrace Coverage
2	Cybersecurity Defense	2-7	Data and Information Protection	2-7-3	The cybersecurity requirements for protecting and handling data and information must include at least the following: 2-7-3-1 Data and information ownership. 2-7-3-2 Data and information classification and labeling mechanisms. 2-7-3-3 Data and information privacy.	Darktrace can be configured to provide more sensitive protection to assets containing sensitive data in order to more aggressively identify and interrupt anomalous activity.
2	Cybersecurity Defense	2-8	Cryptography	2-8-3	The cybersecurity requirements for cryptography must include at least the following: 2-8-3-1 Approved cryptographic solutions standards and its technical and regulatory limitations. 2-8-3-2 Secure management of cryptographic keys during their lifecycles. 2-8-3-3 Encryption of data in-transit and at-rest as per classification and related laws and regulations.	Darktrace can be configured to enforce specific encryption protocols.
2	Cybersecurity Defense	2-9	Backup and Recovery Management	2-9-2	The cybersecurity requirements for backup and recovery management must be implemented.	Through integrations with backup and recovery tools, Darktrace can help security teams understand their states of readiness.
2	Cybersecurity Defense	2-9	Backup and Recovery Management	2-9-3	The cybersecurity requirements for backup and recovery management must include at least the following: 2-9-3-1 Scope and coverage of backups to cover critical technology and information assets. 2-9-3-2 Ability to perform quick recovery of data and systems after cybersecurity incidents. 2-9-3-3 Periodic tests of backup's recovery effectiveness.	Through integrations with backup and recovery tools, Darktrace can help security teams understand their states of readiness.
2	Cybersecurity Defense	2-10	Vulnerabilities Management	2-10-2	The cybersecurity requirements for technical vulnerabilities management must be implemented.	Darktrace continuously monitors assets for known and new vulnerabilities, provides risk-based recommendations for vulnerability remediation, and autonomously implements enhanced detection and response for temporary remediation.
2	Cybersecurity Defense	2-10	Vulnerabilities Management	2-10-3	The cybersecurity requirements for technical vulnerabilities management must include at least the following: 2-10-3-1 Periodic vulnerabilities assessments. 2-10-3-2 Vulnerabilities classification based on criticality level. 2-10-3-3 Vulnerabilities remediation based on classification and associated risk levels. 2-10-3-4 Security patch management. 2-10-3-5 Subscription with authorized and trusted cybersecurity resources for up-to-date information and notifications on technical vulnerabilities.	Darktrace continuously monitors assets for known and new vulnerabilities, provides risk-based recommendations for vulnerability remediation, and autonomously implements enhanced detection and response for temporary remediation.
2	Cybersecurity Defense	2-11	Penetration Testing	2-11-2	The cybersecurity requirements for penetration testing processes must be implemented.	Darktrace augments penetration tests through automation of tasks such as mapping the attack surface and identifying attack paths for testing.

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2	Cybersecurity Defense	2-11	Penetration Testing	2-11-3	The cybersecurity requirements for penetration testing processes must include at least the following: 2-11-3-1 Scope of penetration tests which must cover Internet-facing services and its technical components including infrastructure, websites, web applications, mobile apps, email and remote access. 2-11-3-2 Conducting penetration tests periodically.	Darktrace augments penetration tests through automation of tasks such as mapping the attack surface and identifying attack paths for testing.
2	Cybersecurity Defense	2-12	Cybersecurity Event Logs and Monitoring Management	2-12-2	The cybersecurity requirements for event logs and monitoring management must be implemented.	Darktrace supports the automated collection and analysis of log records.
2	Cybersecurity Defense	2-12	Cybersecurity Event Logs and Monitoring Management	2-12-3	The cybersecurity requirements for event logs and monitoring management must include at least the following: 2-12-3-1 Activation of cybersecurity event logs on critical information assets. 2-12-3-2 Activation of cybersecurity event logs on remote access and privileged user accounts. 2-12-3-3 Identification of required technologies (e.g., SIEM) for cybersecurity event logs collection. 2-12-3-4 Continuous monitoring of cybersecurity events. 2-12-3-5 Retention period for cybersecurity event logs (must be 12 months minimum).	Darktrace supports the automated collection and analysis of log records.
2	Cybersecurity Defense	2-13	Cybersecurity Incident and Threat Management	2-13-2	The requirements for cybersecurity incidents and threat management must be implemented.	Darktrace provides incident response capabilities with varying levels of automation such as playbooks, reporting, and collaboration.
2	Cybersecurity Defense	2-13	Cybersecurity Incident and Threat Management	2-13-3	The requirements for cybersecurity incidents and threat management must include at least the following: 2-13-3-1 Cybersecurity incident response plans and escalation procedures. 2-13-3-2 Cybersecurity incidents classification. 2-13-3-3 Cybersecurity incidents reporting to NCA. 2-13-3-4 Sharing incidents notifications, threat intelligence, breach indicators and reports with NCA. 2-13-3-5 Collecting and handling threat intelligence feeds.	Darktrace provides incident response capabilities with varying levels of automation such as playbooks, reporting, and collaboration.
2	Cybersecurity Defense	2-14	Physical Security	2-14-2	The cybersecurity requirements for physical protection of information and technology assets must be implemented.	Darktrace supports monitoring and reporting on physical protection systems.

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2	Cybersecurity Defense	2-14	Physical Security	2-14-3	The cybersecurity requirements for physical protection of information and technology assets must include at least the following: 2-14-3-1 Authorized access to sensitive areas within the organization (e.g., data center, disaster recovery center, sensitive information processing facilities, security surveillance center, network cabinets). 2-14-3-2 Facility entry/exit records and CCTV monitoring. 2-14-3-3 Protection of facility entry/exit and surveillance records. 2-14-3-4 Secure destruction and re-use of physical assets that hold classified information (including documents and storage media). 2-14-3-5 Security of devices and equipment inside and outside the organization's facilities.	Darktrace supports monitoring and reporting on physical protection systems.
2	Cybersecurity Defense	2-15	Web Application Security	2-15-2	The cybersecurity requirements for external web applications must be implemented.	Darktrace continuously discovers and monitors an organization's externally facing assets and assesses them for risks and vulnerabilities.
2	Cybersecurity Defense	2-15	Web Application Security	2-15-3	The cybersecurity requirements for external web applications must include at least the following: 2-15-3-1 Use of web application firewall. 2-15-3-2 Adoption of the multi-tier architecture principle. 2-15-3-3 Use of secure protocols (e.g., HTTPS). 2-15-3-4 Clarification of the secure usage policy for users. 2-15-3-5 Multi-factor authentication for users' access.	Darktrace continuously discovers and monitors an organization's externally facing assets and assesses them for risks and vulnerabilities.
3	Cybersecurity Resilience	3-1	Cybersecurity Resilience Aspects of Business Continuity Management (BCM)	3-1-2	The cybersecurity requirements for business continuity management must be implemented.	Darktrace's incident simulations assist organizations in defining and testing their recovery plans to limit business impact.
3	Cybersecurity Resilience	3-1	Cybersecurity Resilience Aspects of Business Continuity Management (BCM)	3-1-3	The cybersecurity requirements for business continuity management must include at least the following: 3-1-3-1 Ensuring the continuity of cybersecurity systems and procedures. 3-1-3-2 Developing response plans for cybersecurity incidents that may affect the business continuity. 3-1-3-3 Developing disaster recovery plans.	Darktrace's incident simulations assist organizations in defining and testing their recovery plans to limit business impact.
4	Third-Party and Cloud Computing Cybersecurity	4-2	Cloud Computing and Hosting Cybersecurity	4-2-2	The cybersecurity requirements related to the use of hosting and cloud computing services must be implemented.	Darktrace provides real-time visibility into cloud assets, architectures, users, and permissions, providing protection against cyber-threats, misconfigurations, vulnerabilities, and compliance issues.

Domain No.	Domain	Subdomain No.	Subdomain	Control No.	Control	Darktrace Coverage
4	Third-Party and Cloud Computing Cybersecurity	4-2	Cloud Computing and Hosting Cybersecurity	4-2-3	In line with related and applicable laws and regulations, and in addition to the applicable ECC controls from main domains (1), (2), (3) and subdomain (4-1), the cybersecurity requirements related to the use of hosting and cloud computing services must include at least the following: 4-2-3-1 Classification of data prior to hosting on cloud or hosting services and returning data (in a usable format) upon service completion. 4-2-3-2 Separation of organization's environments (specifically virtual servers) from other environments hosted at the cloud service provider. 4-2-3-3 Organization's information hosting and storage must be inside the Kingdom of Saudi Arabia.	Darktrace provides real-time visibility into cloud assets, architectures, users, and permissions, providing protection against cyber-threats, misconfigurations, vulnerabilities, and compliance issues.
5	Industrial Control Systems Cybersecurity	5-1	Industrial Control Systems (ICS) Protection	5-1-2	The cybersecurity requirements related to Industrial Controls Systems and Operational Technology(ICS/OT) must be implemented.	Darktrace provides real-time visibility into OT networks to identify assets, potential threats, vulnerabilities, and risks. Darktrace's AI can detect and interrupt anomalous activity within OT environments.
5	Industrial Control Systems Cybersecurity	5-1	Industrial Control Systems (ICS) Protection	5-1-3	In addition to the applicable ECC controls from the main domains (1), (2), (3) and (4), the cybersecurity requirements related to Industrial Controls Systems and Operational Technology (ICS/OT) must include at least the following: 5-1-3-1 Strict physical and virtual segmentation when connecting industrial production networks to other networks within the organization (e.g., corporate network). 5-1-3-2 Strict physical and virtual segmentation when connecting systems and industrial networks with external networks (e.g., Internet, wireless, remote access). 5-1-3-3 Continuous monitoring and activation of cybersecurity event logs on the industrial networks and its connections. 5-1-3-4 Isolation of Safety Instrumental Systems (SIS). 5-1-3-5 Strict limitation on the use of external storage media. 5-1-3-6 Strict limitation on connecting mobile devices to industrial production networks. 5-1-3-7 Periodic review and secure configuration and hardening of industrial, automated, support systems, and devices. 5-1-3-8 Vulnerability management for industrial control systems and operational technology (ICS/OT). 5-1-3-9 Patch management for industrial control systems and operational technology (ICS/OT). 5-1-3-10 Cybersecurity applications management related to the protection of the industrial systems from viruses and malware.	Darktrace provides real-time visibility into OT networks to identify assets, potential threats, vulnerabilities, and risks. Darktrace's AI can detect and interrupt anomalous activity within OT environments.